

COLORADO STATE UNIVERSITY PUEBLO

School of Engineering

A Graph Neural Surrogate for Longitudinal Train Dynamics

Committee Progress Update

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Where Things Stand

Thesis chapters

- 1–6 Written; revised this month to the current design
 - 7 First draft: GPU integrator and dataset
 - 8 First draft: surrogate architecture
 - 9–12 Not started: training, evaluation, control, conclusion
-

Built and working

- Reference simulator, route pipeline and RailLab console
- GPU simulator and two 10,000-scenario datasets
- The surrogate, trained and tested over 200-step rollouts

111 pages

Compiled thesis, chapters 1–8

10,000

Scenarios per dataset, 10 to 150
vehicles

Not built

Control demonstration

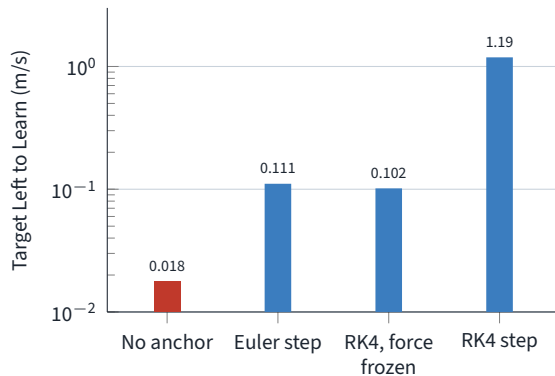
The Surrogate Design Changed

The original plan combined three ideas. One was kept and two were dropped.

Part of the original plan	Status	Reason
Graph neural network: vehicles as nodes, couplers as edges	Kept	Handles any train length with one set of weights.
Learn a correction to the physics equations	Dropped	The training data come from those same equations, so the correction is exactly zero.
Neural ODE: learn a derivative and integrate it with a solver	Dropped	Stiff couplers would force 5 or more network calls per stored step, between states no data can check.

New design: a graph network simulator. The network predicts one fixed 0.25 s step. One network call replaces 50 evaluations of the physics equations, and more of the physics is built in than before, not less.

Why: The Couplers Are Stiff at This Step Size



Median over 480 transitions of data/v2. Lower is an easier learning problem.

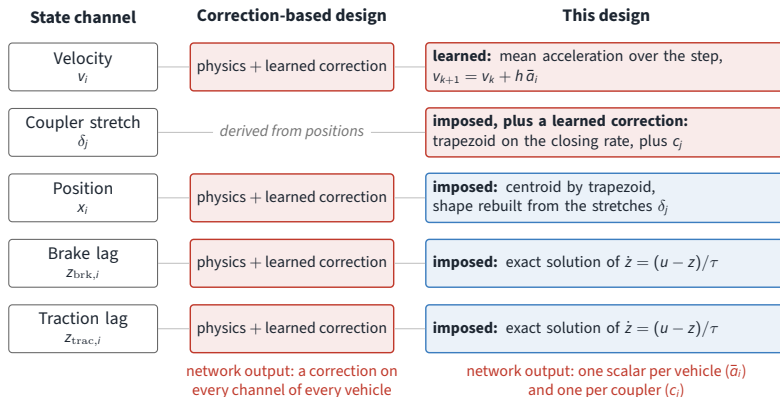
An **anchor** is a cheap physics estimate of the next state; the network would learn only what is left over.

For velocity, **every physics anchor leaves more to learn**, about $6\times$ more at best. One RK4 step of 0.25 s diverges.

The cause is stiffness: a coupler between two heavy cars oscillates with a period as short as 0.21 s, and an explicit step of 0.25 s is unstable for it ($\omega h = 7.45$ against a limit of about 2.8).

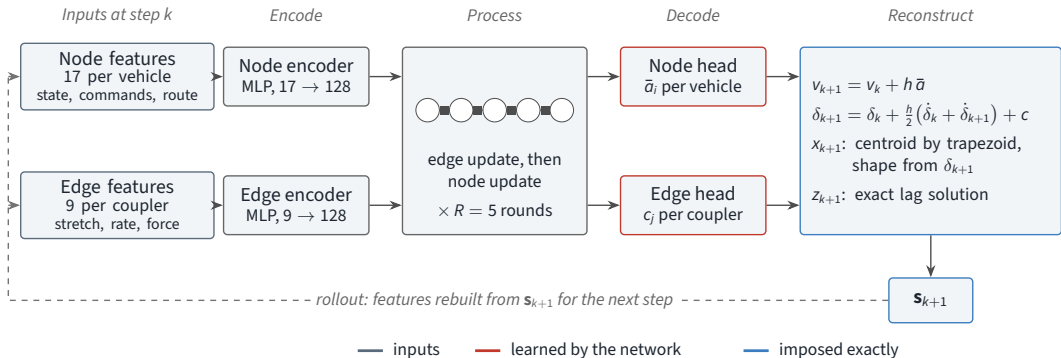
For position the anchor helps: $\dot{x} = v$ exactly, so it is **113 $\times$ better**.

What the Network Predicts, and What Is Computed Exactly



The brake and traction lags have an exact solution, and position follows from velocity. The network predicts **two numbers**: a mean acceleration per vehicle and a shape correction per coupler.

The Network



730,370

parameters, the same for any
train length

5 rounds

set by how far a force wave
travels in 0.25 s

1 call

per 0.25 s step, replacing 50

Data: The GPU Simulator and the Route Fix

GPU simulator

~**170 CPU-h** → **25 min**

Time to build a 10,000-scenario dataset

Flat in N

A 150-car train costs the same as an 11-car one

Agrees with the original simulator within 0.05 m/s
and 5% on peak coupler force.

Route fix

- One of the five corridors had elevation samples that were not on the track: 19.3% of its 10 m steps jumped more than a metre.
- It is now rejected by a check on the raw elevation. Four corridors remain.
- The larger corridor search failed on OpenStreetMap and was switched to the national rail network dataset: 274,145 km of main line.
- **Both datasets were built before this fix**, so they will be rebuilt.

Result: Velocity Holds Up on Longer Trains

Trained on trains of 10 to 80 vehicles. Tested on trains of 121 to 150, which it never saw: 1.5 to 15 times longer.

	Validation, 10–80 cars		Unseen, 121–150 cars	
	Model	vs. baseline	Model	vs. baseline
Velocity error				
One step, RMSE (m/s^2)	0.017	7.8× better	0.020	7.4× better
After 200 steps, 50 s (m/s)	0.066	27× better	0.118	11× better

Baselines: predicting no change (one step) and holding the starting state (200 steps). Stable rollouts needed two training techniques together: noise added to the inputs, and training on the model's own earlier predictions. **Doubling training time nearly halved the 200-step error**, so the model is not yet converged.

Result: Coupler Shape Does Not Hold Up on Longer Trains

Shape error, 200 steps	Model	vs. hold
Validation, 10–80 cars	5.8 mm	6.8× better
Unseen, 121–150 cars	39 mm	1.2× better

Coupler shape sets coupler force, and the safety limit in the control chapter is a limit on coupler force. This is the result that most affects the remaining work.

Three explanations tested and ruled out

- **Not too few message-passing rounds.** Error is even along the train, not larger far from the ends.
- **Not a steady drift.** Removing the drift made results worse on validation and no better on long trains.
- **Not caused by velocity error.** Giving the model perfect velocities removed only 9% of the shape error.

Per-step accuracy is similar on both sets. On long trains the errors **reinforce instead of cancelling**: a stability problem, not an accuracy problem.

Open Problems

- 1 No corridors held out for the control demonstration.** With only four corridors, reserving one more leaves too little to train on.
- 2 The steep-grade test set is built on the rejected corridor,** so it tests false grades rather than real mountain grades.
- 3 The national rail network is not yet split into routes.** It is a graph of 95,936 segments; turning it into corridors is the next piece of code. Everything above waits on it.
- 4 Coupler shape on long trains** (previous slide), which matters most for the control chapter.

Planned fix for 1 and 2: extract corridors including real steep crossings such as Raton Pass, then rebuild with 50,000 to 100,000 scenarios, about 2 to 4 hours on the GPU. Results so far will be re-run on the new data.

Questions for the Committee

- 1 Uncertainty scope.** Uncertainty is now used only as a safety margin on coupler force. Should Chapters 2 and 3 cut the probabilistic forecasting material to match?
- 2 Overspeed runs.** About 18% of scenarios exceed the speed limit by more than 5 m/s, because the simulator does not enforce it. Remove them from training, or keep them as hard cases?
- 3 Control demonstration scope.** Train the control model on all train lengths, which avoids the shape problem, and hold out corridors instead of train lengths?

Next steps, in order

- 1 Split the national rail network into corridors
- 2 Rebuild the dataset on the corrected, larger corridor set
- 3 Retrain and re-run the results
- 4 Build the control demonstration

Target dates: [to be filled in]

Remaining writing

- 9 Training (can be written now)
- 10 Evaluation (after the rebuild)
- 11 Control demonstration
- 12 Discussion and conclusion